

✓ Problem 2 (Regression)

- Import needed libraries
- import csv file and split it into 3 datasets [training , validation , testing]

```
import pandas as pd
import numpy as np
from sklearn.linear_model import LinearRegression, Lasso, Ridge
from sklearn.metrics import mean_squared_error, mean_absolute_error
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv("California_Houses.csv")
```

```
import seaborn as sns
import matplotlib.pyplot as plt

# --- 1. Calculate the correlation matrix from your DataFrame ---
correlation_matrix = df.corr()

# --- 2. Create the heatmap ---
plt.figure(figsize=(12, 10)) # You can adjust the figure size as needed
sns.heatmap(
    correlation_matrix,
    annot=True,           # Show the correlation values on the heatmap
    fmt='.2f',           # Format the numbers to two decimal places
    cmap='coolwarm'      # Use the 'coolwarm' color map
)
plt.title('Correlation Matrix of Features', fontsize=16)
plt.show()
```

Correlation Matrix of Features



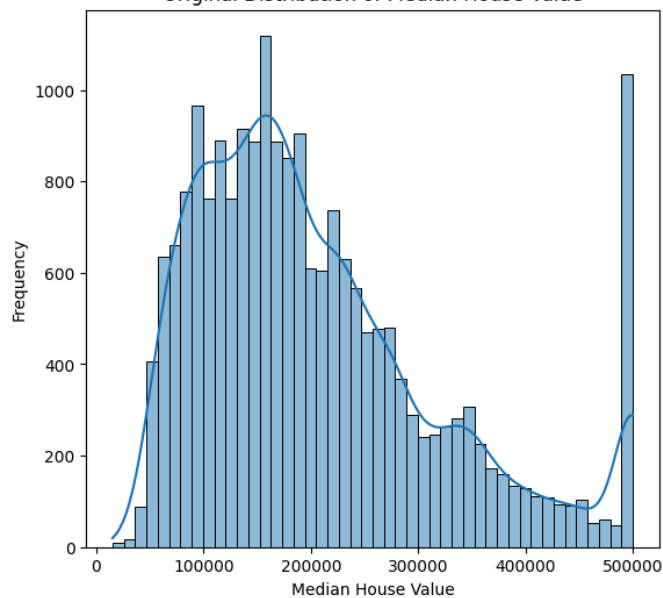
```
# Create a figure for plotting
plt.figure(figsize=(14, 6))

# Plot the original distribution of the target variable
plt.subplot(1, 2, 1)
sns.histplot(df['Median_House_Value'], kde=True)
plt.title('Original Distribution of Median House Value')
plt.xlabel('Median House Value')
plt.ylabel('Frequency')

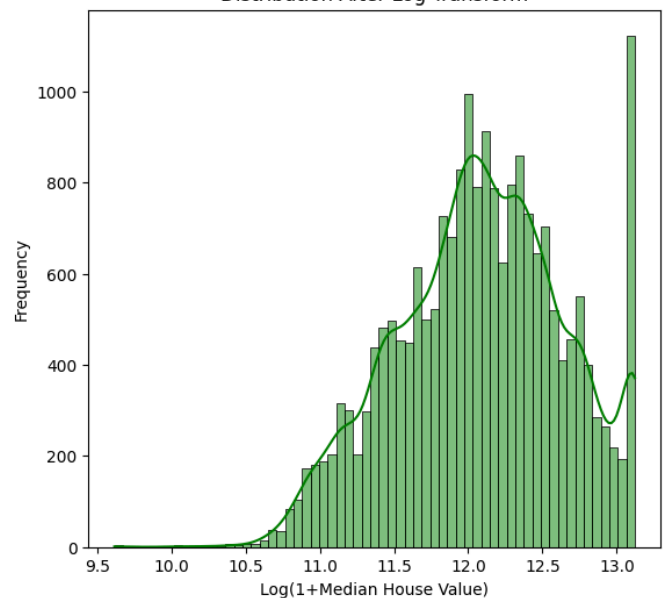
# Apply a log transform to the target variable
# We use np.log1p which calculates log(1 + x) to avoid issues with zero values
df['Median_House_Value_Log'] = np.log1p(df['Median_House_Value'])

# Plot the new distribution after log transform
plt.subplot(1, 2, 2)
sns.histplot(df['Median_House_Value_Log'], kde=True, color='green')
plt.title('Distribution After Log Transform')
plt.xlabel('Log(1+Median House Value)')
plt.ylabel('Frequency')
```

Text(0, 0.5, 'Frequency')
Original Distribution of Median House Value



Distribution After Log Transform



```
df = df.sample(frac=1, random_state=42).reset_index(drop=True)

train_size = int(0.7 * len(df))
```

```

val_size = int(0.15 * len(df))

train_df = df[:train_size]
val_df = df[train_size:train_size + val_size]
test_df = df[train_size + val_size:]

```

calculate mean squared error and mean absolute error

```

def mean_squared(y_true, y_pred):
    return np.mean((y_pred - y_true) ** 2)

def mean_absolute(y_true, y_pred):
    return np.mean(np.abs(y_pred - y_true))

```

Normalize datasets , $X = (X - \text{mean}) / \text{standard deviation}$

```

means = train_df.mean(numeric_only=True)
stds = train_df.std(numeric_only=True)
train_norm = (train_df - means) / stds
val_norm = (val_df - means) / stds
test_norm = (test_df - means) / stds

```

Added Bias column for datasets and dropped target column

```

target_col = "Median_House_Value"

X = train_norm.drop(columns=[target_col]).to_numpy()
V = val_norm.drop(columns=[target_col]).to_numpy()
Test = test_norm.drop(columns=[target_col]).to_numpy()

xones = np.ones((X.shape[0], 1))
vones = np.ones((V.shape[0], 1))
tones = np.ones((Test.shape[0], 1))

X_with_bias = np.hstack((xones, X))
V_with_bias = np.hstack((vones, V))
Test_with_bias = np.hstack((tones, Test))

x_target= train_norm[target_col].to_numpy().reshape(-1, 1)
v_target= val_norm[target_col].to_numpy().reshape(-1, 1)
t_target= test_norm[target_col].to_numpy().reshape(-1, 1)

```

Manual Implementation

Calculate weights using Linear Regression - Direct form

```

W_lin = np.dot(np.linalg.inv(np.dot(X_with_bias.T, X_with_bias)), np.dot(X_with_bias.T, x_target))
print("Closed-form solution Weights:")
print(W_lin)

```

```

Closed-form solution Weights:
[[-2.40784768e-16]
 [ 1.33177124e-01]
 [ 7.43066531e-02]
 [-6.11866752e-02]
 [ 5.60736504e-02]
 [-6.91383126e-02]
 [ 6.96856452e-02]
 [-6.09600918e-01]
 [-3.33871172e-02]
 [ 1.50901509e-01]
 [ 9.13742562e-03]
 [ 6.65991198e-01]
 [ 7.01533443e-03]
 [ 9.86064653e-02]
 [ 9.15364926e-01]]

```

Test weights on validation dataset and calculate MSE and MAE

```
y_pred_dir_form = np.dot(v_with_bias, W_lin)
mse_lin_manual_validation = mean_squared(v_target, y_pred_dir_form)
mae_lin_manual_validation = mean_absolute(v_target, y_pred_dir_form)
print("Mean Squared Error (Validation):", mse_lin_manual_validation)
print("Mean Absolute Error (Validation):", mae_lin_manual_validation)
```

```
Mean Squared Error (Validation): 0.07832257252541956
Mean Absolute Error (Validation): 0.2136381492980262
```

Calculate weights using Linear Regression - Ridge regularization with different lambdas

```
print("\nRidge Regression with Different Lambdas:")
N = X_with_bias.shape[0]
I = np.eye(X_with_bias.shape[1])
lambdas = [0, 0.000001, 0.00001, 0.0001, 0.001, 0.01, 0.1, 1, 10, 100]

results1 = []
results2 = []
for lam in lambdas:
    temp_w_Ridge = np.dot(np.linalg.inv(np.dot(X_with_bias.T, X_with_bias) + lam * N * I), np.dot(X_with_bias.T, x_target))
    y_pred_ridge = np.dot(v_with_bias, temp_w_Ridge)
    mse_ridge_temp = mean_squared(v_target, y_pred_ridge)
    mae_ridge_temp = mean_absolute(v_target, y_pred_ridge)
    results1.append((lam, mse_ridge_temp))
    results2.append((lam, mae_ridge_temp))
    print(f"Lambda = {lam} | Validation MSE = {mse_ridge_temp} | Validation MAE = {mae_ridge_temp}")
best_lambda, best_mse = min(results1, key=lambda x: x[1])
print("\nBest Lambda:", best_lambda)
print("Lowest Validation MSE:", best_mse)
best_lambda, best_mae = min(results2, key=lambda x: x[1])
print("\nBest Lambda:", best_lambda)
print("Lowest Validation MAE:", best_mae)

W_ridge = np.dot(np.linalg.inv(np.dot(X_with_bias.T, X_with_bias) + best_lambda * N * I), np.dot(X_with_bias.T, x_target))
mse_ridge_validation = best_mse
mae_ridge_validation = best_mae
```

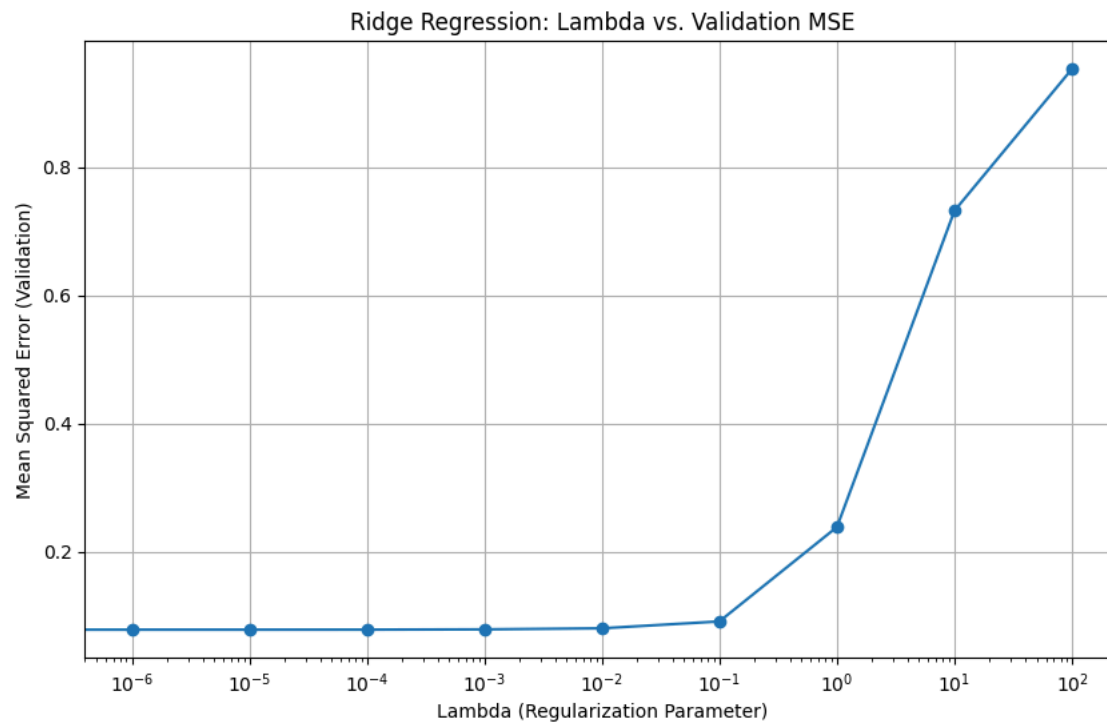
```
Ridge Regression with Different Lambdas:
Lambda = 0 | Validation MSE = 0.07832257252541956 | Validation MAE = 0.2136381492980262
Lambda = 1e-06 | Validation MSE = 0.0783229834725331 | Validation MAE = 0.21363871652588612
Lambda = 1e-05 | Validation MSE = 0.07832670754417391 | Validation MAE = 0.21364420944193618
Lambda = 0.0001 | Validation MSE = 0.0783661705330657 | Validation MAE = 0.21370207705614527
Lambda = 0.001 | Validation MSE = 0.07881949831155163 | Validation MAE = 0.2142043077644963
Lambda = 0.01 | Validation MSE = 0.08047990098498928 | Validation MAE = 0.21546274234492133
Lambda = 0.1 | Validation MSE = 0.09120591752270586 | Validation MAE = 0.22360658440879638
Lambda = 1 | Validation MSE = 0.2384363846300218 | Validation MAE = 0.3607563301236462
Lambda = 10 | Validation MSE = 0.7317023642360704 | Validation MAE = 0.6750932821948713
Lambda = 100 | Validation MSE = 0.9528931954314575 | Validation MAE = 0.7741907289946244

Best Lambda: 0
Lowest Validation MSE: 0.07832257252541956

Best Lambda: 0
Lowest Validation MAE: 0.2136381492980262
```

graph for ridge regression

```
lambdas = [lam for lam, _ in results1]
mses = [mse for _, mse in results1]
plt.figure(figsize=(10, 6))
plt.plot(lambdas, mses, marker='o')
plt.xscale('log')
plt.xlabel('Lambda (Regularization Parameter)')
plt.ylabel('Mean Squared Error (Validation)')
plt.title('Ridge Regression: Lambda vs. Validation MSE')
plt.grid(True)
plt.show()
```



Calculate weights using Linear Regression - Gradient Descent

```
w_gd = np.zeros((X_with_bias.shape[1],1))
learning_rate = 0.01
iterations = 1000
tolerance = 1e-6
prev_mse = float('inf')

for i in range(10000):
    y_pred_training_GD = np.dot(X_with_bias, w_gd)
    errors = y_pred_training_GD - x_target
    gradients = (1/N) * np.dot(X_with_bias.T, errors)
    w_gd -= learning_rate * gradients

    current_mse = mean_squared(x_target, y_pred_training_GD)
    if abs(prev_mse - current_mse) < tolerance:
        print(f"Converged after {i} iterations.")
        break
    prev_mse = current_mse

print("Weights from Gradient Descent:\n", w_gd)
mse_lin_manual_gd_validation = current_mse

y_pred_validation_GD = np.dot(v_with_bias, w_gd)
mae_lin_manual_gd_validation = mean_absolute(v_target, y_pred_validation_GD)
print("Final MSE from Gradient Descent:", mse_lin_manual_gd_validation)
print("Final MAE from Gradient Descent:", mae_lin_manual_gd_validation)
```

```
Converged after 1548 iterations.
Weights from Gradient Descent:
[[ 4.91414666e-16]
 [ 1.35086729e-01]
 [ 7.28687769e-02]
 [-2.80656520e-02]
 [ 6.26541659e-02]
 [-9.09374975e-02]
 [ 5.13406065e-02]
 [-3.47874836e-02]
 [-4.87213891e-02]
 [ 9.63395314e-02]
 [ 7.96021173e-03]
 [ 2.09690514e-02]
 [ 1.40090347e-02]
 [ 1.87377213e-02]
 [ 8.98531337e-01]]
Final MSE from Gradient Descent: 0.08055401806391178
Final MAE from Gradient Descent: 0.21702899255743024
```

```

N = X_with_bias.shape[0]
d = X_with_bias.shape[1]

learning_rate = 0.01
iterations = 10000
tolerance = 1e-6

lambdas = [0, 0.001, 0.01, 0.1, 1, 10]
results = []
best_w = None

for lam in lambdas:
    w = np.zeros((d, 1))
    prev_mse = float('inf')

    for i in range(iterations):
        y_pred_training_ridge_GD = np.dot(X_with_bias, w)
        errors = y_pred_training_ridge_GD - x_target
        gradients = (1/N) * (np.dot(X_with_bias.T, errors) + lam * w)
        w -= learning_rate * gradients
        current_mse = mean_squared(x_target, y_pred_training_ridge_GD)
        if abs(prev_mse - current_mse) < tolerance:
            break
        prev_mse = current_mse

    y_pred_validation_ridge_GD = np.dot(v_with_bias, w)
    mae = mean_absolute(v_target, y_pred_validation_ridge_GD)
    results.append((lam, current_mse, mae, w))
    print(f"Lambda = {lam:<6} | MSE = {current_mse:.6f} | MAE = {mae:.6f}")

best_lambda, best_mse, best_mae, w_ridge_gd = min(results, key=lambda x: x[1])

print("\nBest Lambda:", best_lambda)
print("Lowest Validation MSE:", best_mse)
print("Corresponding MAE:", best_mae)
print("Best Weights:\n", w_ridge_gd)

mse_ridge_gd_validation = best_mse
mae_ridge_gd_validation = best_mae

```

Lambda = 0	MSE = 0.080554	MAE = 0.217029
Lambda = 0.001	MSE = 0.080554	MAE = 0.217029
Lambda = 0.01	MSE = 0.080554	MAE = 0.217029
Lambda = 0.1	MSE = 0.080553	MAE = 0.217029
Lambda = 1	MSE = 0.080554	MAE = 0.217025
Lambda = 10	MSE = 0.080564	MAE = 0.216993

```

Best Lambda: 0.1
Lowest Validation MSE: 0.08055322041049479
Corresponding MAE: 0.21702866919950864
Best Weights:
[[ 4.92041558e-16]
 [ 1.35077018e-01]
 [ 7.28682528e-02]
 [-2.80885051e-02]
 [ 6.26517513e-02]
 [-9.09161378e-02]
 [ 5.13437999e-02]
 [-3.48133929e-02]
 [-4.87349154e-02]
 [ 9.63548739e-02]
 [ 7.98335942e-03]
 [ 2.09766091e-02]
 [ 1.40149150e-02]
 [ 1.87492752e-02]
 [ 8.98550486e-01]]

```

Scikit-Learn Implementation

```

lin_model = LinearRegression(fit_intercept=False)
lin_model.fit(X_with_bias, x_target)
y_pred_lin = lin_model.predict(v_with_bias)

mse_lin_scikit_validation = mean_squared_error(v_target, y_pred_lin)
mae_lin_scikit_validation = mean_absolute_error(v_target, y_pred_lin)

print("Linear Regression:")
print(f"  MSE = {mse_lin_scikit_validation}")
print(f"  MAE = {mae_lin_scikit_validation}")
print("-" * 40)

```

```

Linear Regression:
  MSE = 0.0783225725254196
  MAE = 0.2136381492980261
-----

```

Calculate weights using Linear Regression - Lasso regularization

```

lasso_model = Lasso(alpha=0.001, fit_intercept=False)
lasso_model.fit(X_with_bias, x_target)
y_pred_lasso = lasso_model.predict(v_with_bias)

mse_lasso_validation = mean_squared_error(v_target, y_pred_lasso)
mae_lasso_validation = mean_absolute_error(v_target, y_pred_lasso)

print("Lasso Regression:")
print(f"  MSE = {mse_lasso_validation}")
print(f"  MAE = {mae_lasso_validation}")
print("-" * 40)

```

```

Lasso Regression:
  MSE = 0.07987174686030576
  MAE = 0.21552922634920518
-----

```

Calculate weights using Linear Regression - Ridge regularization

```

ridge_model = Ridge(alpha=1.0, fit_intercept=False)
ridge_model.fit(X_with_bias, x_target)
y_pred_ridge = ridge_model.predict(v_with_bias)

mse_ridge_scikit_validation = mean_squared_error(v_target, y_pred_ridge)
mae_ridge_scikit_validation = mean_absolute_error(v_target, y_pred_ridge)

print("Ridge Regression:")
print(f"  MSE = {mse_ridge_scikit_validation}")
print(f"  MAE = {mae_ridge_scikit_validation}")
print("-" * 40)

```

```

Ridge Regression:
  MSE = 0.07835225950295122
  MAE = 0.21368285800112338
-----

```

Final Results Comparison (validtion dataset)

```

print("*****Linear Regression*****")
print("manual mse:", mse_lin_manual_validation)
print("manual mae:", mae_lin_manual_validation)
print("scikit mse:", mse_lin_scikit_validation)
print("scikit mae:", mae_lin_scikit_validation)
print("-" * 40)
print()

print("*****Linear Regression (Gradient Descent)*****")
print("manual mse (gd):", mse_lin_manual_gd_validation)
print("manual mae (gd):", mae_lin_manual_gd_validation)
print("-" * 40)
print()

print("*****Ridge Regression*****")

```

```

print("manual mse:", mse_ridge_validation)
print("manual mae:", mae_ridge_validation)
print("manual with gradient descent mse:", mse_ridge_gd_validation)
print("manual with gradient descent mae:", mae_ridge_gd_validation)
print("scikit mse:", mse_ridge_scikit_validation)
print("scikit mae:", mae_ridge_scikit_validation)
print("-" * 40)
print()

```

```

print("*****Lasso Regression*****")
print("scikit mse:", mse_lasso_validation)
print("scikit mae:", mae_lasso_validation)

```

```

*****Linear Regression*****
manual mse: 0.07832257252541956
manual mae: 0.2136381492980262
scikit mse: 0.0783225725254196
scikit mae: 0.2136381492980261
-----

```

```

*****Linear Regression (Gradient Descent)*****
manual mse (gd): 0.08055401806391178
manual mae (gd): 0.21702899255743024
-----

```

```

*****Ridge Regression*****
manual mse: 0.07832257252541956
manual mae: 0.2136381492980262
manual with gradient descent mse: 0.08055322041049479
manual with gradient descent mae: 0.21702866919950864
scikit mse: 0.07835225950295122
scikit mae: 0.21368285800112338
-----

```

```

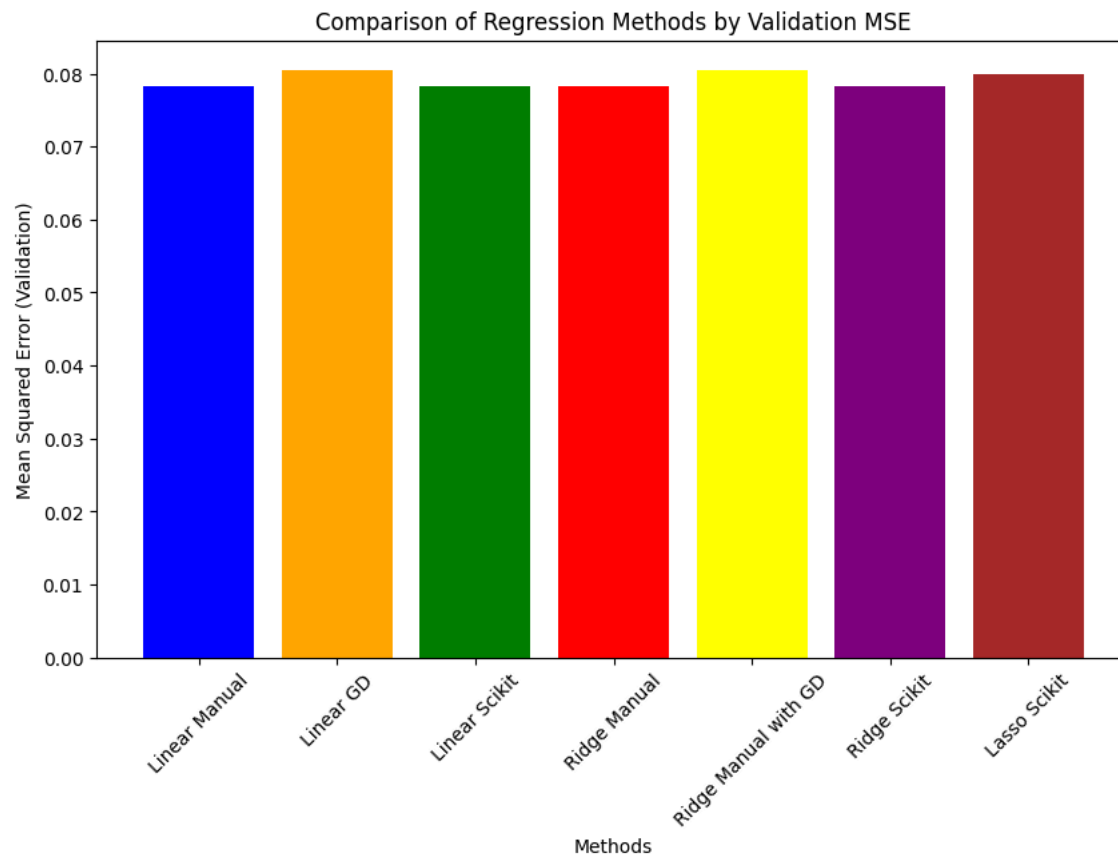
*****Lasso Regression*****
scikit mse: 0.07987174686030576
scikit mae: 0.21552922634920518

```

```

# graph for all methods on validation set
methods = ['Linear Manual', 'Linear GD', 'Linear Scikit', 'Ridge Manual', 'Ridge Manual with GD', 'Ridge Scikit', 'Lasso Scikit']
mses = [mse_lin_manual_validation, mse_lin_manual_gd_validation, mse_lin_scikit_validation, mse_ridge_validation, mse_ridge_gd_vali
plt.figure(figsize=(10, 6))
plt.bar(methods, mses, color=['blue', 'orange', 'green', 'red', 'yellow', 'purple', 'brown'])
plt.xlabel('Methods')
plt.ylabel('Mean Squared Error (Validation)')
plt.title('Comparison of Regression Methods by Validation MSE')
plt.xticks(rotation=45)
plt.show()

```

✓ applying trained models on test dataset & calculating error

```

print("*****Linear Regression*****")
prediction_lin = np.dot(Test_with_bias, W_lin)
mse_lin_manual_test = mean_squared(t_target, prediction_lin)
mae_lin_manual_test = mean_absolute(t_target, prediction_lin)
print(f"Manual Linear Regression on Test Set | Test MSE = {mse_lin_manual_test} | Test MAE = {mae_lin_manual_test}")

prediction_lin_scikit = lin_model.predict(Test_with_bias)
mse_lin_scikit_test = mean_squared_error(t_target, prediction_lin_scikit)
mae_lin_scikit_test = mean_absolute_error(t_target, prediction_lin_scikit)
print(f"Scikit-learn Linear Regression on Test Set | Test MSE = {mse_lin_scikit_test} | Test MAE = {mae_lin_scikit_test}")
print()

print("*****Linear Regression (Gradient Descent)*****")
prediction_gd = np.dot(Test_with_bias, w_gd)
mse_gd_test = mean_squared(t_target, prediction_gd)
mae_gd_test = mean_absolute(t_target, prediction_gd)
print(f"Manual Linear Regression (GD) on Test Set | Test MSE = {mse_gd_test} | Test MAE = {mae_gd_test}")
print()

print("*****Ridge Regression*****")
prediction_ridge = np.dot(Test_with_bias, W_ridge)
mse_ridge_test = mean_squared(t_target, prediction_ridge)
mae_ridge_test = mean_absolute(t_target, prediction_ridge)
print(f"Manual Ridge Regression on Test Set | Test MSE = {mse_ridge_test} | Test MAE = {mae_ridge_test}")

prediction_ridge_gd = np.dot(Test_with_bias, w_ridge_gd)
mse_ridge_gd_test = mean_squared(t_target, prediction_ridge_gd)
mae_ridge_gd_test = mean_absolute(t_target, prediction_ridge_gd)
print(f"Manual Ridge Regression (GD) on Test Set | Test MSE = {mse_ridge_gd_test} | Test MAE = {mae_ridge_gd_test}")

prediction_ridge_scikit = ridge_model.predict(Test_with_bias)
mse_ridge_scikit_test = mean_squared_error(t_target, prediction_ridge_scikit)
mae_ridge_scikit_test = mean_absolute_error(t_target, prediction_ridge_scikit)
print(f"Scikit-learn Ridge Regression on Test Set | Test MSE = {mse_ridge_scikit_test} | Test MAE = {mae_ridge_scikit_test}")

print()

```

```
print("*****Lasso Regression*****")
prediction_lasso = lasso_model.predict(Test_with_bias)
mse_lasso_test = mean_squared_error(t_target, prediction_lasso)
mae_lasso_test = mean_absolute_error(t_target, prediction_lasso)
print(f"Scikit-learn Lasso Regression on Test Set | Test MSE = {mse_lasso_test} | Test MAE = {mae_lasso_test}")
```

*****Linear Regression*****

```
Manual Linear Regression on Test Set | Test MSE = 0.08348359448266321 | Test MAE = 0.2194138730092479
Scikit-learn Linear Regression on Test Set | Test MSE = 0.08348359448266317 | Test MAE = 0.21941387300924778
```

*****Linear Regression (Gradient Descent)*****

```
Manual Linear Regression (GD) on Test Set | Test MSE = 0.0865060532624837 | Test MAE = 0.22165235661199256
```

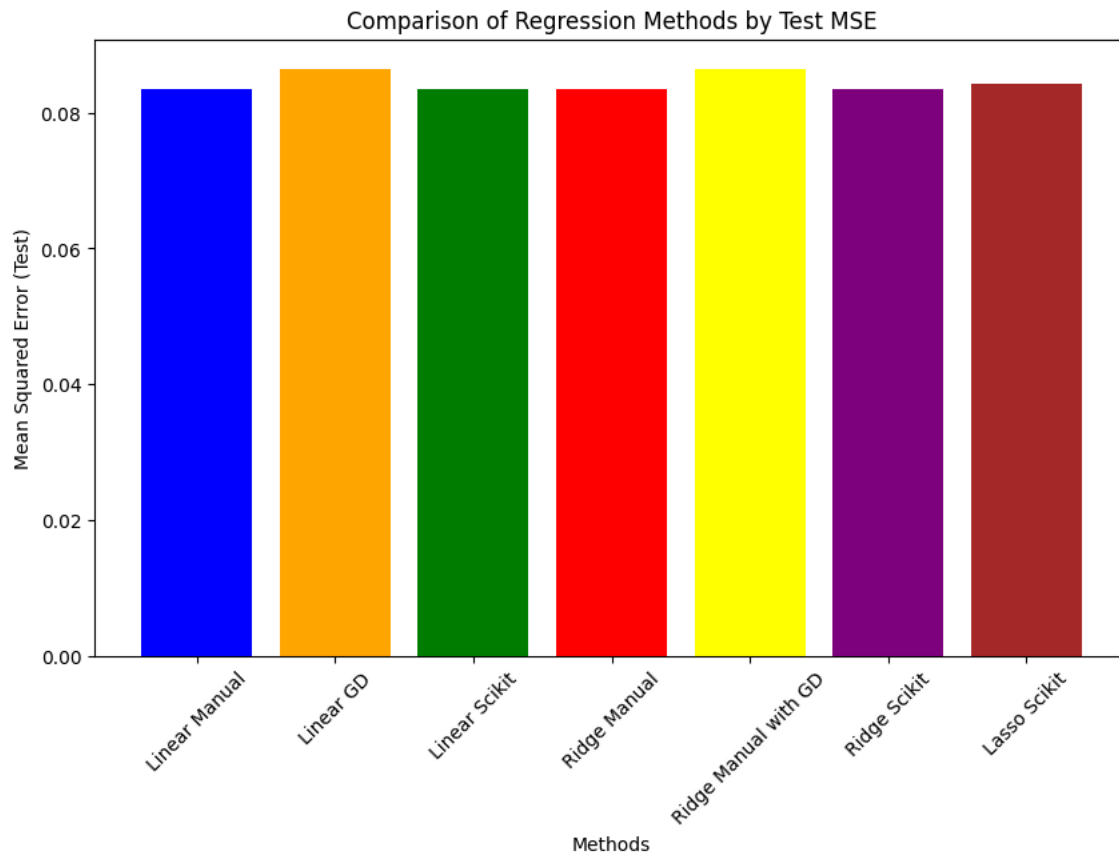
*****Ridge Regression*****

```
Manual Ridge Regression on Test Set | Test MSE = 0.08348359448266321 | Test MAE = 0.2194138730092479
Manual Ridge Regression (GD) on Test Set | Test MSE = 0.08650467291082382 | Test MAE = 0.2216513844756966
Scikit-learn Ridge Regression on Test Set | Test MSE = 0.08347748955532666 | Test MAE = 0.2194117876061641
```

*****Lasso Regression*****

```
Scikit-learn Lasso Regression on Test Set | Test MSE = 0.08426234287069818 | Test MAE = 0.22035400406304614
```

```
# graph for all methods on test set
methods = ['Linear Manual', 'Linear GD', 'Linear Scikit', 'Ridge Manual', 'Ridge Manual with GD', 'Ridge Scikit', 'Lasso Scikit']
mses = [mse_lin_manual_test, mse_gd_test, mse_lin_scikit_test, mse_ridge_test, mse_ridge_gd_test, mse_ridge_scikit_test, mse_lasso_test]
plt.figure(figsize=(10, 6))
plt.bar(methods, mses, color=['blue', 'orange', 'green', 'red', 'yellow', 'purple', 'brown'])
plt.xlabel('Methods')
plt.ylabel('Mean Squared Error (Test)')
plt.title('Comparison of Regression Methods by Test MSE')
plt.xticks(rotation=45)
plt.show()
```



```
methods = [
    'Linear Manual',
    'Linear GD',
    'Linear Scikit',
    'Ridge Manual',
    'Ridge Manual with GD',
    'Ridge Scikit',
    'Lasso Scikit'
]
```

```

mse_test = [
    mse_lin_manual_test,
    mse_gd_test,
    mse_lin_scikit_test,
    mse_ridge_test,
    mse_ridge_gd_test,
    mse_ridge_scikit_test,
    mse_lasso_test
]

mse_validation = [
    mse_lin_manual_validation,
    mse_lin_manual_gd_validation,
    mse_lin_scikit_validation,
    mse_ridge_validation,
    mse_ridge_gd_validation,
    mse_ridge_scikit_validation,
    mse_lasso_validation
]

x = np.arange(len(methods))
width = 0.35

plt.figure(figsize=(12, 6))

plt.bar(x - width/2, mse_validation, width, label='Validation', color='skyblue')
plt.bar(x + width/2, mse_test, width, label='Test', color='salmon')

plt.xlabel('Methods')
plt.ylabel('Mean Squared Error')
plt.title('Comparison of Regression Methods: Validation vs Test MSE')
plt.xticks(x, methods, rotation=45, ha='right')
plt.legend()
plt.tight_layout()
plt.show()

```

